TEXORA · SESSIONAL NOTES · PRACTICAL STUDY SHEET

Analysis of Woven Fabric: General Procedure, Determination of Specifications for Reproduction, and Production Calculations

Given an unlabelled woven sample, extract its complete working specification — weave, thread densities, crimps, yarn counts, weight — then turn the measurements into the numbers a mill needs to reproduce it: reed and draft, warp length, and loom production. Every step is worked on one cotton sample so the arithmetic can be followed and checked line by line.

TE-3206 · FABRIC STRUCTURE AND DESIGN SESSIONAL · STUDY 1 · LEVEL 3 TERM 2

1. Object and order of work

Object: to determine, from a woven sample, every item needed to reproduce it, and to present the result as one specification sheet (Section 9) with its production calculations (Section 8). Work in this order — identification before destruction, cheap tests before yarn is unravelled:

1. **Condition and orient the sample** (Section 2): find the warp direction and the face.
2. **Extract the weave design** on point paper (Section 3).
3. **Count ends and picks** per cm (Section 4).
4. **Measure warp and weft crimp** (Section 5).
5. **Determine both yarn counts** from extracted yarn (Section 6) — its length, corrected by crimp, and its weight.
6. **Weigh the fabric** and check the weight calculated from Sections 4–6 (Section 7).
7. **Write the reproduction specification**: draft, denting and reed (Section 8), warp requirement, and expected loom production.

Condition of the sample. Work on fabric relaxed and flat, away from selvedges, creases and cut edges, at the laboratory's standard atmosphere where counts are critical — the woven counterpart of the relaxation rule of TE-3102/3205.

2. Orientation: warp direction and face

- **Selvedge:** the warp runs parallel to the selvedges; a sample torn (not cut) along a thread line splits along warp or weft cleanly — tear both ways and note which gives the straighter tear.
- **Density and yarn:** the warp is usually the denser, stronger, finer and more highly twisted system, and often sized (a slightly stiffer handle along the warp from residual size).
- **Reed marks and crimp:** fine lengthwise lines (reed marks) run warp-way; the system with the *lesser* crimp is usually the warp in balanced cloths, since warp is held under tension on the loom.
- **Face:** the face shows the clearer design and finish; in faced weaves (TE-3205, Topics 3–4) the dominant system identifies it — a 3/1 face is the warp face. Record face and warp direction on the sample card before any thread is pulled: every later measurement is meaningless if the systems are swapped.

3. Extracting the weave design

Method. Under a counting glass, follow one pick across the sample and record, end by end, whether the end is over or under it; repeat pick by pick (from the bottom of the design upward, TE-3205 Topic 1) until the pattern repeats in both directions. Work on the **face**, with a needle to lift threads, and mark a full repeat plus one extra end and pick to prove the join.

> **RESULT — THE SAMPLE OF THIS SHEET**
>
> The interlacing alternates at every crossing in both directions: the design is **plain weave 1/1**, repeat 2 × 2 (design as TE-3205 Topic 1, Figure 1). No ends or picks work in groups (no rib or hopsack character), so the remaining analysis is of a true plain cloth — the reference case for the whole procedure.

4. Ends and picks per centimetre

Apparatus: counting glass with a centimetre scale, or a travelling microscope for fine cloths. Count over at least 2–3 cm and divide; count at **three places** and average, avoiding the selvedge zone (where density is deliberately higher) and any fault.

> **WORKED RESULT — DENSITIES**
>
> Ends counted over 3 cm at three places: 108, 107, 109 → average 108 ÷ 3 = **36 ends per cm** (= 36 × 2.54 ≈ **91 EPI**).
>
> Picks over 3 cm: 96, 95, 97 → average 96 ÷ 3 = **32 picks per cm** (= 32 × 2.54 ≈ **81 PPI**).

5. Crimp measurement

Method. Mark two points exactly **20 cm** apart along a warp end in the cloth (ink dots or inserted pins). Extract that end, straighten it under just enough tension to remove the bends without stretching, and measure its length. Repeat for a weft pick across a 20 cm width. Crimp % = (yarn length − 20) ÷ 20 × 100.

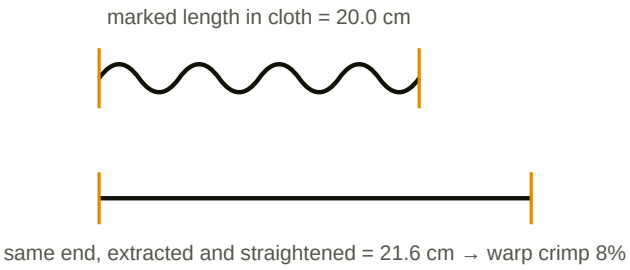


Figure 1 — Crimp is a length difference, nothing more: the yarn a 20 cm strip of cloth consumes, measured after straightening. (Drawing vertically compressed; the wave is exaggerated for visibility.)

WORKED RESULT — CRIMPS

Warp end from the 20 cm mark: straightened length **21.6 cm** → crimp = (21.6 − 20) ÷ 20 × 100 = **8%**.

Weft pick from a 20 cm width: **21.0 cm** → crimp = (21.0 − 20) ÷ 20 × 100 = **5%**.

The warp crimps more here although plain weave is symmetric in design — because the warp was woven under tension and the weft inserted slack, the standard plain-cloth result and a useful orientation check (Section 2).

6. Yarn counts from extracted yarn

Method. Cut a piece exactly **10 cm × 10 cm**. Unravel **all** its warp ends and all its picks. Straighten each system's yarn (as in Section 5) so that total length = threads × straightened length per thread; weigh each system separately. Count = weight and total length, in tex and Ne.

WORKED RESULT — YARN COUNTS (VERIFY EACH LINE)

Warp: the piece holds 36 × 10 = 360 ends; each straightens to 10 × 1.08 = 10.8 cm → total 360 × 10.8 = 3,888 cm = **38.88 m**, weighing **0.574 g**.

tex = 0.574 × 1,000 ÷ 38.88 = **14.76 tex**; Ne = 590.5 ÷ 14.76 = **40 (Ne 40)**.

Weft: the piece holds 32 × 10 = 320 picks; each straightens to 10 × 1.05 = 10.5 cm → total **33.6 m**, weighing **0.551 g**.

tex = 0.551 × 1,000 ÷ 33.6 = **16.4 tex**; Ne = 590.5 ÷ 16.4 = **36 (Ne 36)**.

> Always use the *straightened* (crimp-corrected) length: counting from the 10 cm fabric length instead would overstate the tex by the crimp percentage — 8% coarse on the warp here.

7. Fabric weight — measured and calculated

Measured: the 10 cm × 10 cm piece (100 cm² = 1/100 m²) weighed **1.13 g** → GSM = 1.13 × 100 = **113 g/m²** (average of three pieces in real work).
Calculated from Sections 4–6, per square metre: warp length = 3,600 ends × 1.08 = 3,888 m; weft length = 3,200 picks × 1.05 = 3,360 m.

> **WORKED CROSS-CHECK — GSM**
>
> Warp mass = 3,888 × 14.76 ÷ 1,000 = **57.4 g**; weft mass = 3,360 × 16.4 ÷ 1,000 = **55.1 g**. Calculated GSM = 57.4 + 55.1 = **112.5 g/m²** against 113 measured — agreement within 0.5%, the difference being the sample's moisture regain and any residual size, which the yarn arithmetic does not count. A disagreement beyond a few per cent means an error upstream: re-check the crimp measurement first (it enters twice), then the counts, then the weighing.

8. Reproduction calculations

- **Draft and shafts:** plain weave — straight draft; use **4 shafts** at this density (9 ends per cm of width carried per shaft) rather than the theoretical 2, to spread friction and heald wear; peg plan plain order on shaft pairs (TE-3205, Topic 2).
- **Reed and denting:** 2 ends per dent → reed = 36 ÷ 2 = **18 dents per cm** (45.7 per inch; select the **46 Stockport** reed). Reed width for a 120 cm grey width: allow the standard contraction margin used by the mill and record it in the specification.
- **Total ends:** 36 × 120 = **4,320 ends** for the body, plus selvedge ends (e.g. 2 × 12 = 24) → **4,344 ends** to be warped.
- **Warp length:** for 100 m of cloth, warp consumed = 100 × 1.08 = **108 m**; add loom waste allowance (say 1.5 m per beam) → prepare ≈ **109.5 m** per end on the beam.
- **Loom production:** at 220 picks per minute and 81.3 picks per inch: picks per hour = 220 × 60 = 13,200; cloth length = 13,200 ÷ 81.3 = 162.4 inches/h = **4.12 m/h at 100% efficiency**; at a realistic 88% efficiency → **3.63 m/h**. (Weft consumed per hour follows from pick length = reed width + fringe allowance, times picks — set it out the same way when costing.)

9. The specification sheet — the finished result

ITEM	RESULT FOR THE SAMPLE OF THIS SHEET

Weave	Plain 1/1 (design extracted, repeat 2 × 2)
Ends / picks	36 per cm (91 EPI) / 32 per cm (81 PPI), averages of three counts
Crimp	Warp 8% (21.6 cm from 20 cm) / weft 5% (21.0 cm)
Warp count	14.76 tex = Ne 40 (38.88 m extracted, 0.574 g)
Weft count	16.4 tex = Ne 36 (33.6 m extracted, 0.551 g)
Fabric weight	113 g/m² measured; 112.5 calculated — agreement within 0.5%
Reproduction	Straight draft on 4 shafts; 2 ends/dent in a 46 Stockport reed (18 dents/cm); 4,344 ends for 120 cm width incl. selvedges; warp 109.5 m per 100 m cloth; expected production 3.63 m/h at 220 ppm, 88% efficiency
Fibre	By burning test and microscopy: cotton (state your own observations for your sample)

10. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. A warp end marked over 25 cm of cloth straightens to 27.0 cm. State its crimp, and why the count calculation must use 27.0 cm rather than 25 cm.
2. From a 10 × 10 cm piece: warp yarn totals 38.88 m and 0.574 g; weft 33.6 m and 0.551 g. Give both counts in tex and Ne.
3. Your calculated GSM is 98 but the pieces weigh out at 113 g/m². Name the two measurements you re-check first, and the likely error in each.
4. Why 4 shafts for a 2-shaft weave at 36 ends/cm — and what goes wrong with 2?

Answer key

1. Crimp = (27.0 − 25) ÷ 25 × 100 = **8%**. Count is mass per unit length of *yarn*; the yarn in 25 cm of cloth is 27.0 cm long, and using 25 cm would understate its length and overstate its tex by 8%.
2. Warp: 0.574 × 1,000 ÷ 38.88 = **14.76 tex** = 590.5 ÷ 14.76 ≈ **Ne 40**. Weft: 0.551 × 1,000 ÷ 33.6 = **16.4 tex** = **Ne 36**.
3. (i) **Crimp** — it enters the yarn length of both systems and then again in the weight build-up; using fabric length instead of straightened length understates both masses (a 13% shortfall is exactly the signature of ignoring crimp). (ii) **Thread densities** — counts taken near a selvedge or over too short a gauge length distort ends/picks per cm, which scale the whole calculation.

4. Load and friction: at 36 ends/cm on 2 shafts, each shaft carries 18 ends per cm of width — heald crowding, high friction, more end-breaks and a cloudy shed. Spreading the same straight-draft order over 4 shafts (ends dealt 1-2-3-4 cyclically, lifts paired) quarters nothing structurally but halves the per-shaft density; the weave is unchanged because shafts 1,3 lift together and 2,4 together.

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